Question 1 - Solution

1)

For a system with $m_1 = m_2 = m$ and spring constants $k_A = k_B = k_C = k$, the equations of motion are derived as:

$$m\ddot{x}_1 = -kx_1 + k(x_2 - x_1) = -2kx_1 + kx_2 \tag{1}$$
$$m\ddot{x}_2 = -kx_2 - k(x_2 - x_1) = -2kx_2 + kx_1 \tag{2}$$

We assume a mode solution of the form $\vec{x} = \mathrm{Re}\left[\begin{pmatrix} A \\ B \end{pmatrix} e^{-i\omega t}\right]$. Substituting $\ddot{x} = -\omega^2 x$ into the equations yields:

$$-\omega^2 mA = -2kA + kB \tag{3}$$
$$-\omega^2 mB = -2kB + kA \tag{4}$$

Defining $\omega_0^2 = k/m$ and $\alpha \equiv (\omega/\omega_0)^2$, we obtain the following linear system:

$$(2 - \alpha)A - B = 0 \tag{5}$$
$$-A + (2 - \alpha)B = 0 \tag{6}$$

In matrix form:

$$\begin{pmatrix} 2-\alpha & -1 \\ -1 & 2-\alpha \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} = 0 \tag{7}$$

For a non-trivial solution, the determinant must be zero:

$$(2 - \alpha)^2 - 1 = 0 \implies 2 - \alpha = \pm 1 \tag{8}$$

The eigenfrequencies are:

- **Slow mode:** $\alpha_s = 1 \implies \omega_s = \omega_0 = \sqrt{k/m}$
- **Fast mode:** $\alpha_f = 3 \implies \omega_f = \sqrt{3}\omega_0$

The eigenvectors are determined by substituting α back:

- For $\alpha_s = 1$: $(2-1)A - B = 0 \implies A = B \implies \vec{v}_s = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$
- For $\alpha_f = 3$: $(2-3)A - B = 0 \implies A = -B \implies \vec{v}_f = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

2)

Consider a system with modes $\vec{v}_s = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\vec{v}_f = \begin{pmatrix} 1 \\ -3 \end{pmatrix}$. Given initial conditions $\vec{x}(0) = \begin{pmatrix} 2a \\ a \end{pmatrix}$ and $\dot{\vec{x}}(0) = \vec{0}$.

Given the symmetry of the initial conditions, it is clear without the use of equations that only the slow mode is energized. The initial conditions are revived.

The general position is $\vec{x}(t) = a_s \cos(\omega_s t + \phi_s)\vec{v}_s + a_f \cos(\omega_f t + \phi_f)\vec{v}_f$. Setting $\dot{\vec{x}}(0) = 0$ implies $\phi_s = \phi_f = 0$. Substituting $t = 0$:

$$\begin{pmatrix} 2a \\ a \end{pmatrix} = a_s \begin{pmatrix} 2 \\ 1 \end{pmatrix} + a_f \begin{pmatrix} 1 \\ -3 \end{pmatrix} \tag{9}$$

By observation, $a_s = a$ and $a_f = 0$. Since only the slow mode is energized, after a full period of the fast frequency $t = \frac{2\pi}{\omega_s}$, the state is:

$$\vec{x}\left(\frac{2\pi}{\omega_s}\right) = a \cos\left(\frac{2\pi\omega_s}{\omega_s}\right)\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} a \tag{10}$$

$$\vec{v}\left(\frac{2\pi}{\omega_s}\right) = -a\omega_s \sin\left(\frac{2\pi\omega_s}{\omega_s}\right)\begin{pmatrix} 2 \\ 1 \end{pmatrix} = \vec{0} \tag{11}$$

3)

Damped oscillator:

$$x\left(t\right) = Ae^{-\gamma t/2}\cos\left(\omega_u t + \phi\right)$$

Define:

$$\tau = \frac{2}{\gamma} \quad T_u = \frac{2\pi}{\omega_u}$$

Ratio:

$$\frac{T}{\tau} = \frac{\pi\gamma}{\omega_u} = \frac{\pi\gamma}{\omega_0\sqrt{1 - \frac{\gamma^2}{2\omega_0^2}}}$$

Under-damped frequency approximation:

$$\gamma = 10^{-2}\omega_0$$

$$\omega_u = \omega_0\sqrt{1 - \frac{\gamma^2}{2\omega_0^2}} \approx \omega_0\left(1 + \frac{\gamma^2}{8\omega_0^2}\right)\omega_0\left(1 + \frac{10^{-4}}{8}\right) \approx \omega_0$$

$$\frac{T}{\tau} \approx \frac{\pi\gamma}{\omega_0} = \frac{\pi 10^{-2}\omega_0}{\omega_0} = \pi 10^{-2}$$

$$\# = \frac{100}{\pi} \approx 32$$

Energy:

$$E_{in} = \frac{k}{2}A^2 = \frac{m\omega_0^2}{2}A^2$$

$$E_f = \frac{m\omega_0^2}{2}A^2/e^2$$

$$\Delta E = \frac{m\omega_0^2}{2}A^2\left(1 - \frac{1}{e^2}\right)$$

Relative part:

$$\frac{\Delta E}{E_{in}} = 1 - \frac{1}{e^2} \approx 0.87$$

4)

At resonance the driving is perfectly in phase the with the oscillation of the mass. As the frequency increases the oscillator phase 'slips' away from the driving force. One may say that the oscillator "can't keep up" with the driving, or that it doesn't have time to respond.